



UTM
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SECJ1023 – PROGRAMMING TECHNIQUE II

SECTION 2

PROJECT 1

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Introduction

The purpose of this proposal is to recommend the development of a Library System. This system includes Object-Oriented Programming (OOP) principles such as encapsulation, inheritance, polymorphism and associations. Besides, this system will be designed to solve the problems faced by students and library administrators in managing the library records.

The Library System allows the user to log in as either the students or the library administrators. Students can search the name of the books that they are interested in through this system and the details of the books such as the book's author and the availability status of the books will be displayed. Next, this system allows students to borrow books if the books are available and return books they have borrowed. Not only that, this system also enables the library administrators to view the fine status of the students and add new library books. Overall, the Library System that will be developed in a C++ environment will help the students and the library administrators to manage the library records efficiently.

Solution

The proposed system will be built using the C++ programming language and will include various functions that will streamline library operations.

The system will include user authentication, which is a login system that ensures only authorised users have access to the system. Book class will contain attributes such as title, author, and availability status. It also includes features for Library class such as title searches and admin-added book additions. For user roles, the Student class allows students to borrow and return books. The Administrator class allows administrators to check on student penalties and add new library materials. There is also a Fine class that handles fines for late or non-returned books. The class will compute and display fees based on the number of days late returned.

Storyboard

Login page and main menu

```
*****
                                LIBRARY SYSTEM
*****

                                =  MAIN MENU  =

                                SELECT OPTIONS
                                -----
                                1. ADMIN    2. STUDENT

                                (PRESS 3 TO EXIT)
```

When u key in 1 for admin option:

(Check student penalty)

```
1

Good Day!!!
Please enter your Admin ID:UTM1234

Please enter your password:mike890

WELCOME ,ADMIN! :) :)
Press 'p' to check student penalties or 'a' to add new books: p
Please enter Student's ID to check his or her penalties : A23CS0001

There is a late fee of RM 25 for this student.
```

(Add new books)

```
1

Good Day!!!
Please enter your Admin ID:jjn

Please enter your password:dv

WELCOME ,ADMIN! :) :)
Press 'p' to check student penalties or 'a' to add new books: a

Enter the name of the book: Forest
Enter the author of the book: Benjamin.JK
Is the book available for students to borrow? (y/n): y

///Book added successfully.///
Add more books? (y/n): |
```

When u key in 2 for student option:

```
*****
                                LIBRARY SYSTEM
*****

                                =  MAIN MENU  =

                                SELECT OPTIONS
                                -----
                                1. ADMIN    2. STUDENT

                                (PRESS 3 TO EXIT)

2

Good Day!!!
Please enter your Student ID:A23CS0001

Please enter your password:hellobello

WELCOME ,STUDENT!   ^_^ ^_^
Click 'r' if you want to return a book, click 'b' if you want to borrow a book ?
```

Click b for borrow book

```
b

PLEASE ENTER THE TITLE OF THE BOOK:How to improve my programming skill

**  BOOK FOUND  :**

Title       : How To Improve My Programming Skill
Author      : Calynn.L
Book Status : Available

Enter 'y' if you want to borrow: y

/// You successfully borrowed the book ///
```

Click r for returning book

```
r

PLEASE ENTER THE TITLE OF THE BOOK:How to improve my programming skill
**  BOOK FOUND  :**

Title       : How To Improve My Programming Skill
Author      : Calynn.L
Book Status : Available

Enter 'y' if you want to return: y

## You successfully returned the book ##
```

When u key in numbers other than 1 and 2:

```
3  
THANK YOU, HAVE A NICE DAY  ^_^  
-----
```

Conclusion

The suggested library system intends to address the issues that students and librarians have in properly maintaining library records. Using OOP concepts, the system will give an organised and flexible solution that can be upgraded in the future. Overall, the construction of the Library System in C++ will help simplify library operations, enhance record administration, and increase the library's overall efficiency.